

Control Systems Study Notes (EE302)

Academic Insight - Public Sample Revision Summary

Module 1: Closed-Loop Transfer Functions & Block Diagrams

- Negative feedback transfer function: $T(s) = G(s) / (1 + G(s) H(s))$
- Characteristic equation: $1 + G(s) H(s) = 0$ governs closed-loop poles

Module 2: Second-Order Transient Response & Routh-Hurwitz

- Standard 2nd-order form: $T(s) = \omega_n^2 / (s^2 + 2\zeta\omega_n s + \omega_n^2)$
- Settling time (2%): $t_s = 4 / (\zeta * \omega_n)$
- Routh-Hurwitz: First-column sign changes equal right-half plane poles.
- Note: Sample educational revision sheet provided by Academic Insight.